The Elbow Method

Determining the Optimal Number of Clusters in K-Means

What is the Elbow Method?

In K-Means clustering, the algorithm requires you to specify the number of clusters (K) beforehand. However, looking at high-dimensional data, it is impossible to visually guess the correct number of clusters. The **Elbow Method** is a heuristic used to determine this optimal value for K.

The Core Metric: WCSS

To use the Elbow Method, we first need to understand a metric called **WCSS** (Within-Cluster Sum of Squares).

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WCSS = Sum of the squared distances between each data point and its corresponding cluster centroid.
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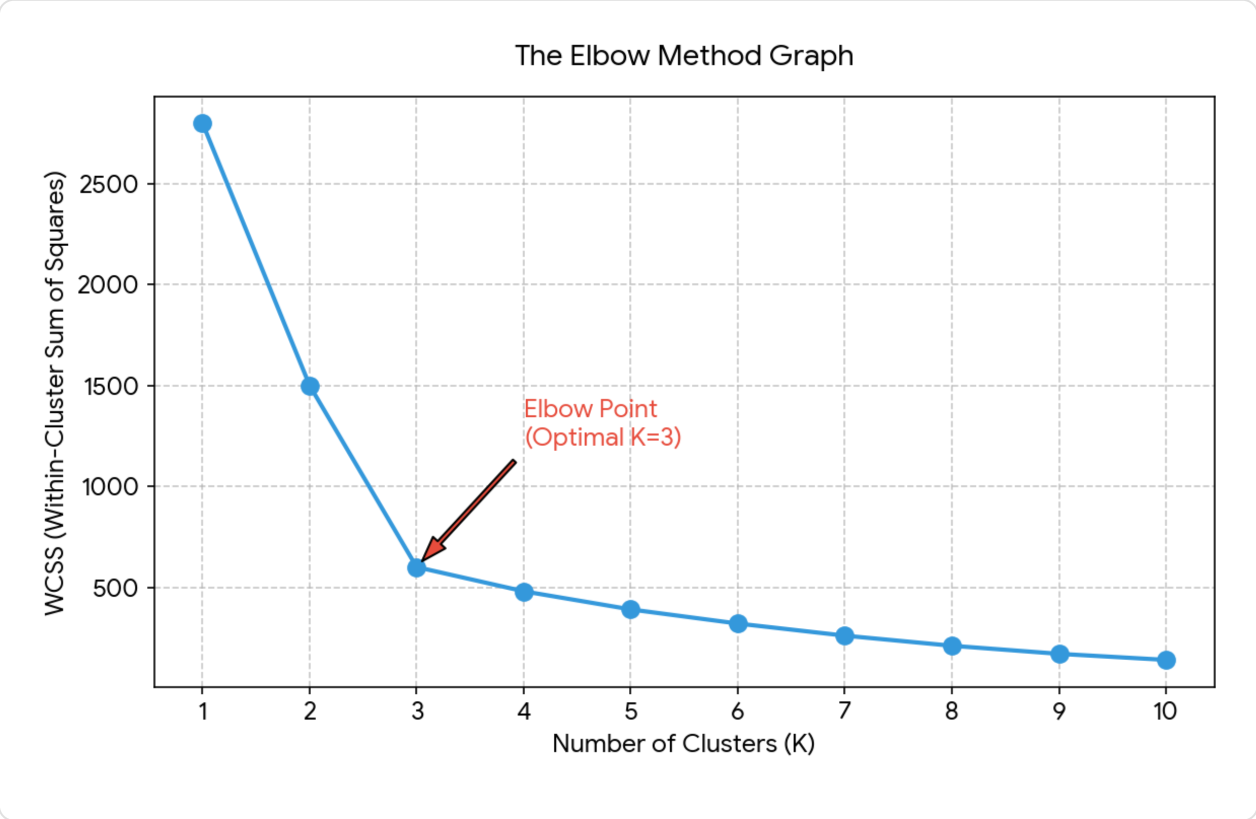
When K=1 (all points in a single cluster), WCSS is at its maximum. As you increase the number of clusters (K), the WCSS value naturally decreases because data points are getting closer to their respective cluster centers. If K equals the total number of data points, WCSS becomes 0, but that defeats the purpose of clustering.

Visualizing the "Elbow" Point

To find the optimal K, we run the algorithm for multiple values of K and plot the resulting WCSS values on a line graph. When you look at the plotted graph, it will resemble an arm.



Initially, as K increases from 1 to 2 or 3, the WCSS drops dramatically. But at a certain point, the rate of decrease drops significantly, and the curve begins to flatten out, looking like the bend of an elbow.

The Optimal K: The point where this sharp drop ends and the line starts to level off is considered the "Elbow Point." In the diagram above, this occurs at **K=3**. Adding more clusters beyond this point does not yield a significant enough reduction in WCSS to justify the computational complexity of additional clusters.